

The problem: The right tool for the wrong job

Introduction

You now understand what agents are and how they work. But here's the big question: when should you actually use them?

Not every problem needs an agent. Sometimes a simple chatbot is enough. Sometimes basic automation does the job. And sometimes, agents are the only solution that makes sense. Knowing the difference will save you from both over-engineering simple problems and under-engineering complex ones.

In this module, we'll explore real-world scenarios where agents shine, situations where they are not a good fit, and how to make the right architectural decision for your use case.



The problem: The right tool for the wrong job

Here's a story that plays out every day. A team gets excited about agents. They've seen the demos, read the papers, and they're ready to build. They pick their first project: an FAQ bot for their website.

Six weeks later, they've built an agent that can reason about questions, search multiple data sources, and compose answers dynamically. It's technically impressive. It's also an overly complex solution for answering questions like, "What are your business hours?" and "How do I reset my password?"

Meanwhile, another team is drowning in complexity. They're trying to build a system that handles customer returns—checking order history, evaluating return policies, calculating refunds, updating inventory, and issuing credits. They're doing it with traditional if-then automation and the code is becoming unmaintainable. This is exactly where an agent would shine, but they don't realize it.

The problem isn't the technology—it's knowing when to use it.

